ARTEFACT · SOFTWARE ENGINEER (SENIOR, BRAZIL)

Artefact written case — Task Manager

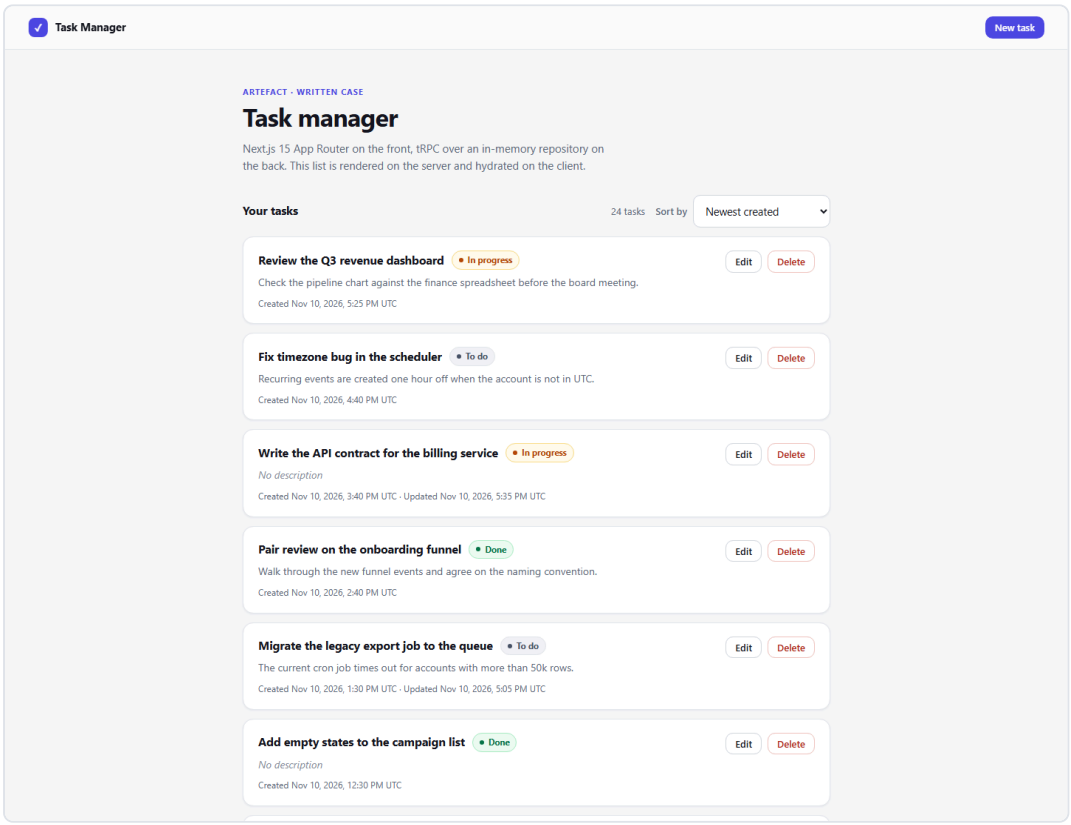
Paulo Cezario · github.com/phscezario/artefact-selective-process · November 10, 2026

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A small task manager built for the **Software Engineer (Senior, Brazil)** written case: **Next.js 15 App Router** on the front, **tRPC** over an **in-memory repository** on the back, with server-side rendering on the list page and an infinite-scroll bonus.

> The full written answers (theory + practical) are in `docs/answers`, together with the two submission PDFs. `npm run submission` rebuilds both from this README and those answers.



Stack

Layer	Choice	Version
Framework	Next.js (App Router)	15.5.29
UI	React / React DOM	19.3
API	tRPC + TanStack React Query integration	11.27 / 5.107
Validation	Zod (shared by server and form)	4.6
Serialization	superjson (keeps `Date` intact on the wire and across RSC)	2.2
Tests	Vitest	3.2
Language	TypeScript (strict)	5.9
Monorepo	npm workspaces	npm 10+

How to run

```
# Node 22+ (see .nvmrc)
npm install
npm run dev          # http://localhost:3000
```

Production mode:

```
npm run build
npm start
```

Docker:

```
docker compose up --build   # http://localhost:3000
```

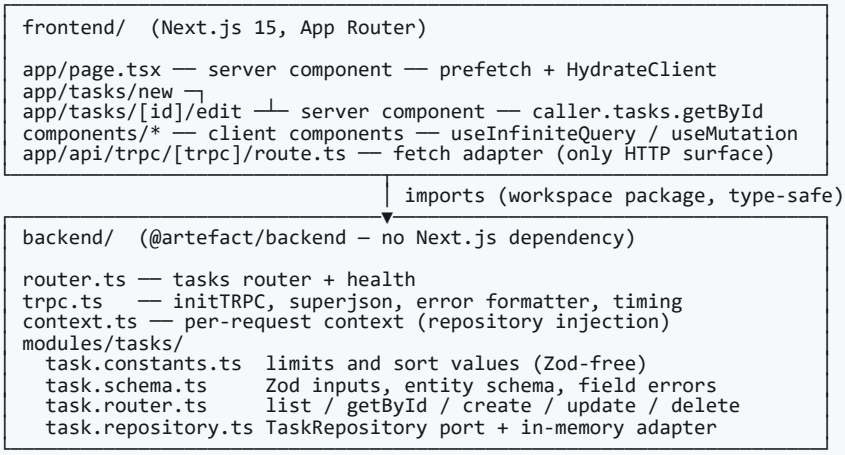
Quality gate (the same command CI runs):

```
npm run verify       # lint -> format:check -> typecheck -> test -> build
```

Submission documents (needs Chrome or Edge installed):

```
npm run submission   # docs/answers + README -> PDFs (submission/ and docs/answers/)
```

Architecture



The backend is a **package, not a process**: the router has no knowledge of Next.js and receives its repository through the context. Mounting it on Fastify/Express later is a ~20-line adapter, and the server components call it **in-process** — no HTTP round trip to your own API on SSR.

API

Single endpoint: `POST|GET /api/trpc/*` (fetch adapter). Procedures:

Procedure	Type	Input	Errors
`tasks.list`	query	`{ limit?: 1..50 (default 8), cursor?: string, sort? }`	`BAD_REQUEST`
`tasks.getById`	query	`{ id }`	`NOT_FOUND`
`tasks.create`	mutation	`{ titulo, descricao?, status? }`	`BAD_REQUEST`
`tasks.update`	mutation	`{ id, titulo?, descricao?, status? }` (at least one field)	`BAD_REQUEST`, `NOT_FOUND`
`tasks.delete`	mutation	`{ id }`	`NOT_FOUND`
`health`	query	—	—

Field names (`titulo`, `descricao`, `dataCriacao`, `dataAtualizacao`) follow the case specification; everything else in the codebase is in English. `dataCriacao` and `dataAtualizacao` are real `Date` objects on both sides (superjson). A task starts with both timestamps equal, and every update bumps `dataAtualizacao` without touching `dataCriacao`.

`sort` accepts `created-desc` (default), `created-asc`, `updated-desc`, `titulo-asc` and `status-asc`; an unsupported value is a `BAD_REQUEST`. Every order is a total order with the task id as tiebreaker, so two pages never disagree about the position of equal timestamps.

Pagination is **keyset**: `nextCursor` is the id of the last item, and the next page starts after it. With `offset/take`, deleting a row between two requests would skip an item; with a cursor the only consequence is that a deleted cursor restarts from the first page.

What is implemented

- **SSR list** — `app/page.tsx` prefetches the first page in a server component, dehydrates the cache and hydrates the client component. `dynamic = 'force-dynamic'` because the in-memory store changes per request (otherwise Next would serve build-time HTML).
- **Ordering** — creation (both directions), last update, title and status. The order lives in the URL (`/?sort=updated-desc`), so the server prefetches exactly that view, the page is shareable and the change is a `startTransition` navigation instead of a client-side reshuffle.
- **Infinite scroll** — TanStack `useInfiniteQuery` + `IntersectionObserver` sentinel with `rootMargin: 240px`; the sentinel only exists while `hasNextPage` is true.
- **Delete with feedback** — inline two-step confirmation (no `window.confirm`), per-row pending state, success/error toasts, and query invalidation of every cached page.
- **Create / edit** — plain hooks (`useTaskForm`), validation with the same Zod schema the API uses, server-side field errors merged back into the form, loading state on submit, redirect + toast.
- **Errors** — `NOT_FOUND` for missing tasks, `BAD_REQUEST` with `fieldErrors` for invalid input, stacks stripped in production, `TRPCClientError` normalised into one readable message.
- **States** — loading skeleton, empty state, error state with retry, end-of-list marker.
- **Seed** — 24 fictional tasks so pagination and the infinite scroll are visible immediately.

Decisions and trade-offs

Decision	Why	Cost
Backend as a workspace package instead of a second server	One command to run, no CORS, SSR calls the router in-process, but the domain still has a hard boundary	Backend isn't independently deployable *today* (the router already is transport-agnostic)
tRPC instead of REST	End-to-end types with zero codegen; the same Zod schema validates the form and the API	Not consumable by non-TypeScript clients; one URL makes per-resource CDN caching unnatural
Zod as the single source of validation	No rule drift between client and server	The schema ships to the client (~13 KB gzipped)
superjson	`dataCriacao` stays a `Date` instead of a stringly-typed field	Small payload overhead
Keyset pagination	Correct under concurrent deletes	Cursor is opaque to the UI
Sort per request	One comparator covers every order, no extra indexes	Editing an item between two pages can move it out of the current cursor window
In-memory repository behind an interface	The case does not need a database; the domain does not care either	Data resets on restart, single instance only
Hand-written CSS	No build step or extra dependency for a case this size	No utility classes for future screens
`npm` workspaces with TypeScript sources	No build/watch step for the backend package	Consumers need `transpilePackages` (configured)

Tests

```
npm test        # vitest run in both workspaces
```

33 backend tests. Twenty-nine of them drive the real router with an injected repository (no HTTP, no mocks of the code under test); four go through the same `fetchRequestHandler` adapter the app mounts, which is the only way to verify what the wire actually promises: `dataCriacao` and `dataAtualizacao` as `Date` objects (superjson), `fieldErrors` on `BAD_REQUEST`, and a production `INTERNAL_SERVER_ERROR` that leaks neither message nor stack. They cover creation with trimmed title and normalised description, both length boundaries, an unknown status, keyset pagination with ordering and no duplicates, an unknown cursor, the default page size, partial updates, the update timestamp, every sort order, `NOT_FOUND` on update/delete/getById, defensive copies, and the 24 seeded tasks.

13 frontend tests cover the pure logic behind the list, the form and the error handling: the sort parser (including unknown values in the URL), the shared list input, the field-error mapping and message extraction from a `TRPCClientError`.

The production build was also smoke-tested end to end (create → update → list → delete → double delete → empty title), including the SSR HTML assertion that seeded tasks are present in the first response.

What I would do next

Persistence (Postgres + a second `TaskRepository`), authentication in `createContext` (the HTTP context already carries the request headers; the RSC caller would resolve them with `next/headers`), per-procedure rate limiting, optimistic updates on delete, Playwright for the critical flow, OpenTelemetry spans instead of the timing `console.info`, and a `/tasks/[id]` detail route.

Repository layout

```
backend/    domain + tRPC router (no Next.js dependency)
frontend/   Next.js app: route handler, server components, client components
docs/
  answers/          written answers (Part 1 and Part 2) + the submission PDFs
  screenshots/      images used by the README and by the PDFs
scripts/            submission PDF builder (Markdown -> HTML -> Chrome headless)
submission/         generated PDFs + screenshots (gitignored)
```